

LESSON 105 (0.5 GROUND and 1.3 DUAL)

Lesson Objective

During this lesson, the student will conduct a preflight inspection in preparation for flight and apply the techniques practiced in the training device in the aircraft. The applicant will increase their understanding of the effects of primary and secondary aircraft controls.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review maneuvers and procedures to be performed in flight.
- Review homework assigned in the previous lesson.
 - Quiz the student on primary and secondary flight controls.
 - Review the passenger briefing, engine fire during start briefing, and the takeoff briefing.
 - Quiz the student on the readings as needed.
 - Review chapter 03 of the Airplane Flying Handbook - Basic Flight Maneuvers:
 - The four fundamentals.
 - Effects and use of flight controls.
 - Attitude flying.
 - Straight and level flight.
- Student questions.

References

- Advisory Circulars. ([Search Tool](#))
 - Pilots' Role in Collision Avoidance. ([AC 90-48E](#))
- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 03 - Basic flight maneuvers.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 02 - Aeronautical Decision-Making.
 - Chapter 06 - Flight Controls.
- Warrior III Pilot's Information Manual.
 - Section 06 - Weight and Balance.
 - Section 07 - Description and operation of the airplane and its systems.

Instructor Guidance

- This is the student's first actual flight — acknowledge that milestone. A brief word of encouragement before departure goes a long way in building confidence and setting a positive tone.
- Manage expectations early. Let the student know that the aircraft will feel different from the simulator — control pressures, environmental factors, and workload will all be new. Normalize

mistakes and frame the lesson as exploration, not evaluation.

- During the preflight inspection, resist the urge to lead. Supervise and prompt rather than demonstrate — the student should be building independence with the checklist from day one.
- Pay close attention to the student's go/no-go decision process. This is their first time applying it in a real-world context, so walk through each element with them rather than simply approving the flight yourself.
- On the ground and during taxi, keep your verbal input minimal but purposeful — students on their first flight are easily overwhelmed. Prioritize the most critical items and save secondary details for the debrief.
- In the air, let the student feel the controls before correcting. Give them a moment to experience cause and effect — this is far more effective than immediate intervention for non-safety-critical deviations.
- Trim is almost always neglected on a student's first flight. If you notice them muscling the aircraft, pause other instruction and refocus on the "pitch to select, trim to relieve" habit before moving on.
- Walk the student through how to log their flight and ensure the student understands how to complete their own logbook entry — this is their first entry and sets the standard going forward.

Lesson Content

Review Weather Briefing

- Quiz the student on:
 - Official vs non-official weather sources.
 - Different ways to obtain a weather briefing.
 - Types of weather briefings.
- Have the student brief the current weather pertaining to the flight.

Review Performance & Weight and Balance (TOLD card)

- Review and inspect the student's TOLD card.
- Ask the student questions on:
 - Weight and balance.
 - Pressure altitude and density altitude.
 - Takeoff and landing distance.
 - V-speeds.

Primary and Secondary Aircraft Controls

- Quiz the student on primary and secondary aircraft controls.

Effects of Flight Controls

- Primary:
 - Elevator → Pitch.

- Rudder → Yaw.
- Ailerons → Roll.
- Secondary:
 - Elevator → Airspeed.
 - Rudder → Roll.
 - Ailerons → Yaw.
- The speed that air is moving over a flight control surface will greatly impact the effectiveness of said surface.
 - Faster airflow will result in more responsive flight control.
 - Slower airflow will result in less responsive flight control.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic when weather permits.
- FAR regulation 91.111 - operating near other aircraft.
 - Cannot operate near another aircraft to create a collision hazard.
 - Cannot fly in formation unless there was a prior agreement between the PIC's of participating aircraft.
 - No passengers for hire in formation flight.
- Explain how to scan for traffic in daytime conditions.
 - 10 degrees per second, looking both above and below aircraft level.
- Clearing procedures:
 - During taxi.
 - Before climbs.
 - During descents.
 - Clearing turns.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Preflight Inspection

- For the first flight, it is recommended that the instructor supervises the student's preflight.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.

- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be considered even once airborne.

Lesson Content (Dual)

Checklist Procedures

- Re-emphasize the importance of using checklists.
- Have the student run through all the checklists.
- Assist if needed.
- Have the student go through the “passenger briefing” and the “engine fire during start brief”.

Engine Handling

- Highlight the red line on the RPM gauge.
- Ensure that the student understands the negative effects of “redlining” the engine.
- When stationary on the ground, RPM should be set to 1000 RPM.
- Do not lean the mixture below 3,000 ft.
 - For leaning guidance, refer to the PIM and the leaning guidance on the FIT hub.
- Operation of Mixture Control
 - The mixture lever has a red knob and is located to the right of the throttle lever.
 - Moving the mixture lever forward results in a rich fuel/air mixture, while moving the mixture lever backward results in a lean fuel/air mixture.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command’s responsibility to avoid traffic.
- Demonstrate how to scan traffic in daytime conditions.
 - 10 degrees per second, looking both above and below aircraft level.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Primary Effects of Flying Controls

- Emphasize to the student that all control movements must be smooth!
- Elevator → Pitch.
- Rudder → Yaw.
- Ailerons → Roll.

Secondary Effects of Flying Controls

- Elevator → Airspeed.
- Rudder → Roll.
- Ailerons → Yaw.
- Introduce the concept of a spiral dive and how to properly recover.

Effects of Airspeed

- The speed that air is moving over a flight control surface will greatly impact the effectiveness of said surface.
 - Faster airflow will result in more responsive flight control.
 - Slower airflow will result in less responsive flight control.

Effect of Slipstream

- Fast.
 - Elevator → More Pitch.
 - Rudder → More Yaw.
 - Ailerons → More Roll.
- Slow.
 - Elevator → Less Pitch.
 - Rudder → Less Yaw.
 - Ailerons → Less Roll.

Effect of Trim

- Emphasize the importance of trim!
- Natural sense.
 - Roll wheel forward for forward trim.
 - Roll wheel aft for aft trim.
- Coarse versus fine trim.
- Pitch to select attitude, trim to relieve pressure.
- Any adjustment to pitch or power should be followed with an adjustment to trim.

Effect of Power and Airspeed:

- For an increase in power.
 - The pitch will increase and there will be a yaw to the left.
 - To correct you need forward pressure on the yoke and right rudder.
- For a decrease in power.

- The pitch will decrease and there will be a yaw to the right.
- To correct you need back pressure on the yoke and left rudder.
- For an increase in airspeed.
 - The pitch will increase and there will be a yaw to the right.
 - To correct you need forward pressure on the yoke and left rudder.
- For a decrease in airspeed.
 - The pitch will decrease and there will be a yaw to the left.
 - To correct you need back pressure on the yoke and right rudder.

Effect of Flaps

- Demonstrate the effects of flaps 10, 25 and 40 degrees.

Adding Flaps:

- Lift → Increases.
- Nose → Goes up.
- Elevator (initially) → Forward.
- Drag → Increases.
- Airspeed → Decreases.
- Elevator (due to drag) → Backward.

Removing Flaps:

- Lift → Decreases.
- Nose → Goes down.
- Elevator (initially) → Backward.
- Drag → Decreases.
- Airspeed → Increases.
- Elevator (due to drag) → Forward.

Operation of Carburetor Heat

- The carburetor heat lever is located to the right of the mixture lever.
- To operate, pull the lever downward to the on position or upward to the off position.
- Review when carburetor heat is required (show in checklists).

Operation of Environmental Controls

- Show where air vents are located in the aircraft, and how to operate.
- Demonstrate how to operate the cabin heat and defroster.

Other Controls as Applicable

- Adjusting audio panel power selection, volume, squelch, Garmin430 basic usage (com

selection, frequency changes, power selection).

Postflight Procedures

- For the first flight, it is recommended that the instructor supervises the postflight procedures.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 106.
- Assign homework:
 - Review Chapter 04 of the Airplane Flying Handbook – Principles of Flight
 - Review Chapter 05 of the Airplane Flying Handbook – Aerodynamics of Flight

Completion Standards

This lesson is complete when the student can:

1. Before the flight:
 - a. Have a weather briefing and completed TOLD card prepared.
2. During the flight:
 - a. Demonstrate an understanding of checklist procedures.
 - b. Demonstrate the effects of primary and secondary controls in the airplane with minimal instructor guidance.